Computational Coupling — Executive Summary

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Four choices decide whether a connectome beats its rewiring.

Computational coupling is the extent to which what a network computes is attributable to how it is wired. It is never measured directly — only as the gap between a network's score and the scores of surrogate networks, under some unit model, at some operating point, through some readout. This project shows that each of those four choices can flip the sign of that gap on identical data, so a coupling reported as a single signed number is under-determined by its own methods section.

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This document replaces a withdrawn one. Version 4 of this project circulated a pitch whose hardware figures — a TSMC 28 nm node, a die area, 116.71 pJ/SynOp, a 455× energy reduction, a "physical tapeout" — were never measured by any tool. They came from a Python model with those values written in as constants. Every one is retracted. The retracted document is kept at `paper/superseded_v4/pitch/` behind its own notice, and `AUDIT.md` gives the evidence for each withdrawal.

This version claims nothing that is not produced by a script in this repository from a result file also in this repository.

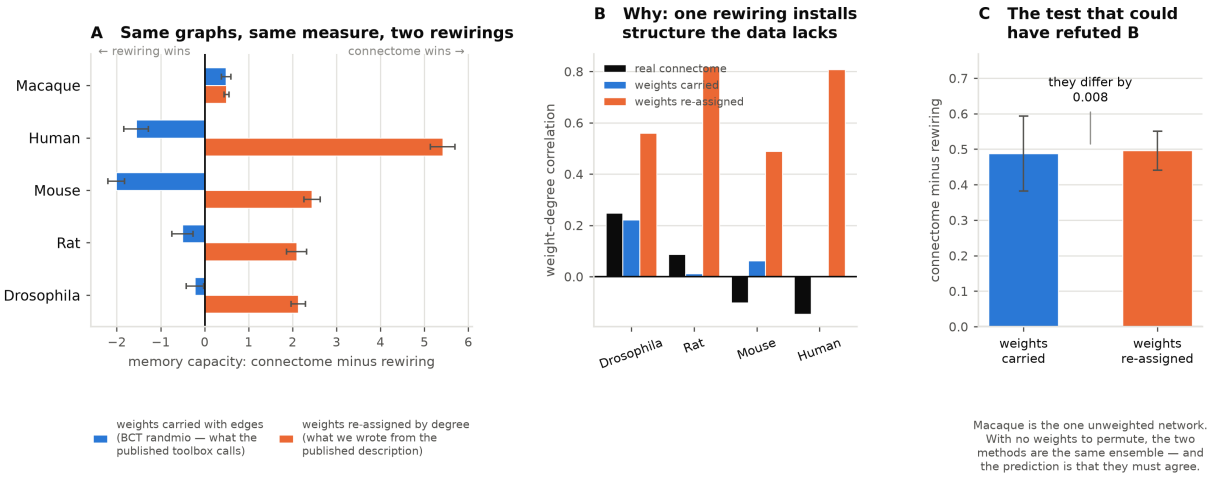
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0. The headline

The reservoir-computing literature reports that empirical connectomes outperform degree-preserving rewirings on memory capacity. That result is increasingly cited as a reason to build **neuromorphic hardware** on connectome-derived wiring.

We tried to check it and found that the comparison is not one experiment. It is a composition of four choices, each normally reported in a subordinate clause, each of which can flip the sign of the answer

on identical data. We measured all four on the five connectomes distributed with the original authors' own toolbox.



(A) The result. (B) The cause: one rewiring installs a weight-degree association the real graphs do not have. (C) The test that could have refuted (B) — on the one network with no weight variation, the two procedures must agree, and they do, to 0.008.

The first one caught us. "Degree-preserving randomisation" names at least two different ensembles — one where each weight travels with its edge (what the published toolbox actually calls), one where the weight multiset is re-assigned to the rewired graph afterwards. Both match the phrase. We built the second believing it was the first:

null ensemble	connectome beats it, graded units	median difference
BCT `randmio` — the published one	**1 of 5** lead, **3 of 5** trail	−0.51
rank-reassigned — ours, by mistake	**5 of 5** lead, all significant	+2.13

Across 10 (network, unit model) cells the two nulls give opposite **signs** in 6 and opposite **significant verdicts** in 3 — one says the connectome wins at $p < 0.05$, the other says it loses at $p < 0.05$, same graphs, same operating point.

Our version produced a clean, large, highly significant result in the expected direction. Nothing in its output indicated it was answering a different question. It was caught by reading the toolbox source, not by anything in the numbers.

We can say why, and the explanation was falsifiable. Rank-reassignment does not merely randomise — it *installs* a weight–degree association the data does not have (+0.66 median shift, against +0.06 for the published ensemble). That predicts the two must agree exactly on a network with no weight variation. One of the five is binary. There they agree to **0.008 capacity units**.

This is not a claim that the published result is wrong — our harness differs in other respects too. It is a demonstration that a replication which builds its null from a published description is not a replication, and cannot tell which hypothesis it tested.

The other three choices: the unit model (graded vs spiking — capacity compresses 5×, and neuromorphic hardware is the spiking one), the operating point (the contrast changes sign along the gain axis in 4 of 5 networks, significant on both sides), and the readout protocol (under an all-to-all readout, a reservoir with its recurrent matrix **set to zero** scores 0.596 against the connectome's 0.596 — the wiring is not being measured at all).

One mechanism we proposed changed sign when the null was corrected, and we report that too. We expected a threshold to silence a heavy-tailed graph's low-degree neurons differentially. Against our mistaken null the silencing gap did not predict the capacity gap, and we wrote the prediction up as failed. Against the published null it does — $r = -0.58$, $p = 2.5\times10^{-5}$. A wrong control manufactured a clean positive result in one place and destroyed a real one in another, from the same code.

1. The problem

Connectomes contain statistically overrepresented subgraphs — feedforward loops, reciprocal inhibitory pairs. A large literature argues these *motifs* are what make biological wiring computationally effective, and a neuromorphic literature builds architectures on that premise.

The argument has never been properly tested, for a reason that is methodological rather than empirical. Establishing that a structural property **causes** a computational advantage needs a control that differs in that property **and nothing else**. The standard comparison — a connectome against a degree-preserving rewiring, or against an Erdős–Rényi graph — does not have that form. Rewiring a connectome changes its motif census *together with* its clustering, its spectrum, and every higher-order statistic that co-varies with them. A performance difference cannot be attributed to motifs.

The control that would settle it does not exist: an ensemble that randomises a connectome while **holding its motif census fixed**.

2. What we built

A motif-preserving null ensemble. Degree-, valence- and census-matched surrogates, sampled by an annealed Metropolis chain over exact, incrementally updated motif counts. On *C. elegans* it reproduces all six targeted motif counts to within **0.3%** while retaining only **13.9%** edge overlap with the empirical graph — against **11.6%** for an unconstrained rewiring. It converges on all five networks we test, including diffusion tractography, where almost every edge is reciprocal and an ordinary swap chain cannot hold the census at all.

A five-rung control ladder. Density → degree sequence → valence placement → motif census → the empirical connectome. Each rung adds exactly one structural property, so each property's contribution is the difference between adjacent rungs, with node count, edge count, spectral radius and input projection matched throughout.

Two controls that catch silent failures. Both are cheap, both are transferable, and both changed our own conclusions:

Control	What it exposes
Zero-recurrence	Under an all-to-all protocol, a reservoir with its recurrent matrix **set to zero** scores 59.6% — agai
Matched firing rate	Matching spectral radius does not match activity level. Erdős–Rényi surrogates run **1.77**× hotter

A gap-free perturbation bound. The Davis–Kahan theorem conventionally invoked for degeneracy is inapplicable to measured connectomes — their spectral gaps make it vacuous. We bound the finite-horizon input–output map instead, which needs no gap, and extract the **operational alignment** γ of a perturbation.

3. What we found

Evaluated as fixed reservoirs on Spiking Heidelberg Digits, across five networks from four species:

Every structural property that measurably moves accuracy moves it down.

Property added	Significant	Median effect
Degree sequence	3 / 5 networks	**−4.65 pp**
Motif census	2 / 5 networks	**−3.50 pp**

Property added	Significant	Median effect
Valence placement	1 / 5 networks	−0.82 pp
Residual wiring	2 / 5 networks	+1.06 pp

Adding a connectome's exact 3-node motif spectrum to a degree- and valence-matched surrogate **costs** accuracy, by about as much as the degree sequence does. That is the opposite sign to the one the motif hypothesis predicts. Task-independent dynamical probes (memory capacity, kernel rank) reproduce the ordering and the direction.

Recomputing enrichment itself, with 500 surrogates and exact counts, already qualifies the premise: of 29 interpretable tests, 23 motifs are enriched, **3 are significantly depleted** — coherent feedforward loops in the *Drosophila* central complex, $Z = -10.9$ — and 3 are indistinguishable from the null. Mouse V1 sits at its null throughout. Motif enrichment is real but not conserved across phylogeny.

4. Why this is worth something

The operating-point result generalises to a whole class of published comparison. Any study that compares a network against a null at one dynamical regime — and most do — inherits a dependence it has not reported. The fix is one sweep rather than one evaluation. That is cheap, and it converts a claim about a point into a claim about a regime.

The null ensemble is a reusable instrument, though not a new idea. Motif-constrained ensembles are standard in statistical network science (ERGMs, the dk-series, and Fischer et al.'s Wang–Landau sampler, which has a sampling guarantee ours does not). What we supply is the sign-attributed version at connectome scale and its use as a *control rung*. Any field that infers function from subgraph overrepresentation faces the same attribution problem — gene regulatory networks, food webs, software dependency graphs.

The controls generalise further than the finding. The zero-recurrence control is four lines and would invalidate a category of published comparison. We would encourage it in any study comparing connectome-derived reservoirs, and we report it on our own work first.

A negative result, properly bounded, is worth more than an unbounded positive one. We state what the design could have detected (80% power at $d = 1.54$ after correction), we state where our own prediction failed, and we state where two of our measurements disagree rather than choosing the one that reads better.

5. What we are not claiming

- **Not** that connectome wiring is bad for computation. It is a poor fixed reservoir for one classification task with a linear readout, which is not the objective any nervous system is under selection for.
- **Not** that motif structure is functionally irrelevant. Our design is powered for $d = 1.54$; it does not exclude moderate effects, and it does not test motifs as an inductive bias for **learning**, which is a different claim requiring a trained ladder we have not run to publication standard.
- **Not** any silicon result beyond what an open flow establishes. We report gate count, flip-flop count and mapped pre-layout cell area from Yosys/Nangate45 — 590,116 cells, 25,644 flip-flops, 749,263 μm^2 for the top macro. **No die area, no power, no energy per operation, no clock frequency.** The toolchain performs no place-and-route and ships no static timing analyser.

6. Where our own prediction failed

We predicted tolerance to structural damage would scale as $1/\gamma$. Tested two ways:

- **Ordering — holds.** γ ranks the perturbation directions correctly in 3 of 5 networks. Tolerated pruning falls from **83%** for random removal to **64%** along the adversarial direction at identical cardinality.
- **Scaling — fails.** The product $\gamma \cdot \|E\|_{max}$ should be constant across directions. It varies by a factor of **2.2**.

We report the failure. γ identifies which direction is dangerous and does not predict by how much. The conclusion that matters for the literature needs only the ordering: robustness to *random* ablation is much weaker evidence of degeneracy than it appears, because a random direction is nearly orthogonal to any fixed functional subspace.

7. Verification

Everything in the manuscript regenerates from raw result files:

```
./src/analysis/rebuild_paper.sh
```

It reports unresolved numbers, undefined references and LaTeX errors, because a clean exit status is not evidence of anything — `pdflatex` exits 0 on a hard error by design. Current state: **0 unresolved, 0 errors, 0 undefined references, 108 tests passing.**

No number in the manuscript is transcribed by hand. A missing result appears in the PDF as a visible placeholder rather than as a stale value.

8. The record

`AUDIT.md` documents 34 entries: **15** defects in the retracted v4 manuscript, **12** found in the **replacement** analysis while building it — our own, recorded on the same terms — and **7** where a correction changed a conclusion, including our own partly falsified prediction and two conclusions that reversed outright when we found that our null ensemble was not the published one.

One pattern runs through nearly all of them, and it is the most transferable thing in this project:

> **Every defect produced a plausible number and no error.** A loader that swallowed its exception returned an empty graph. An `argsort` over equal values returned index order. A null model imposed a constraint it did not impose. A protocol control decoded nothing and scored chance. `pdflatex` reported a duplicate macro and exited 0 with a complete PDF. None was caught by a result looking wrong. Each was caught by checking a recorded diagnostic, or by reading the code against the prose.

That is why the convergence diagnostics, the protocol control and the held-out statistic are reported in the paper rather than kept in the repository.

Repository: analysis code, null-model implementations, experiment drivers, the synthesis flow, the audit and the retracted v4 material. **Manuscript:** `paper/computational_coupling_v5.pdf` (27 pages). **Full technical detail:** `info.md`.